

Paper Summary: **Pantheon360**

Taming Digital Twin Generation via 3D-Aware 360° Video Diffusion

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Project page: https://koi953215.github.io/pantheon360_page/ Paper (CVF):
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<https://arxiv.org/abs/2605.25449>

Abstract

Generating complete digital twins from videos requires precise camera control, global scene coverage, and strict spatial-temporal consistency, constraints that remain challenging for perspective video generators due to their limited field of view (FoV). Their narrow FoV forces long or multi-view trajectories, amplifying cross-view inconsistency and temporal drift. The authors argue that 360° video generation offers a natural solution: panoramic coverage simplifies trajectory design and provides strong global context for maintaining coherence. They introduce **Pantheon360**, a controllable 360° video generation framework that synthesizes high-fidelity video from sparse 360° inputs. The key idea is an explicit *3D Cache*, reconstructed from the input, which serves as a geometric scaffold for any user-defined camera path, allowing the diffusion model to focus on photorealistic texture refinement while the 3D Cache enforces global geometric consistency. Experiments show Pantheon360 achieves superior visual quality and unmatched geometric coherence, enabling reliable, flexible 360° scene generation for downstream simulation and digital-twin applications.

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1 Problem and Motivation

Digital twin generation for robotics and autonomous-agent simulation traditionally relies on static 3D reconstruction (NeRF, 3D Gaussian Splatting), which cannot easily produce dynamic, photorealistic worlds with low human effort. Generative video models are a promising alternative but face two hard problems: (1) 3D-aware controllability and (2) long-term temporal consistency.

The dominant paradigm, *camera-controlled perspective video generation*, is fundamentally limited because a narrow field of view leaves the model blind to most of the scene from the initial frame. For long or multi-trajectory exploration, the model must repeatedly hallucinate unseen regions, causing redundant conditioning (re-observing the same geometry from different views) and severe spatial/temporal inconsistency as the generated world contradicts itself.

Key insight: 360° video captures the entire scene context from $t = 0$, giving a holistic understanding that perspective models lack. This simplifies trajectory representation and improves coherence. However, 360° generation introduces its own challenges: extreme equirectangular-projection (ERP) distortion and, most critically, the difficulty of *precise* geometric camera control. Prior controllable 360° models are limited:

- **GenEx** supports only simple, high-level actions (e.g., “move forward”), not exact trajectory following.
- **CamPVG** achieves precise trajectory control but is validated only on synthetic data, leaving in-the-wild applicability unproven.

2 Method

Pantheon360 is built on a pre-trained latent video diffusion model, Stable Video Diffusion (SVD), fine-tuned with an explicit 3D-aware conditioning mechanism.

2.1 Problem Formulation

Given sparse 360° input frames $\{I_k\}$ and a target camera trajectory $C_{target} = \{c_1, \dots, c_T\}$, the goal is to generate a temporally consistent 360° video $Y_{equi} \in \mathbb{R}^{T \times 3 \times H' \times W'}$ in equirectangular format. Two elements drive the approach: an explicit **3D Cache** for geometric conditioning, and **360° video generation** for global consistency.

2.2 3D Cache Reconstruction

At inference time, each 360° input frame is cropped into multiple perspective views and passed to a 3D reconstruction foundation model (e.g., π^3 or VGGT) to produce a 3D point cloud that explicitly models the scene’s spherical geometry. The framework is agnostic to the specific reconstruction backbone (also compatible with CUT3R, DUST3R, MAST3R-style methods).

2.3 Geometric Conditioning (V_{geo})

Given the target trajectory C_{target} , the reconstructed point cloud is rendered into equirectangular format along that trajectory, producing a *geometry-only* video

$$V_{geo} \in \mathbb{R}^{T \times 3 \times H' \times W'}.$$

V_{geo} is passed through the SVD VAE encoder $\mathcal{E}$ to obtain a latent scaffold

$$v_{equi} = \mathcal{E}(V_{geo}),$$

which is concatenated with the noised latent at every diffusion step, injecting precise geometric structure into the denoising process.

2.4 Model Architecture

The generator G is a fine-tuned SVD U-Net f_θ (using the pretrained SVD VAE encoder $\mathcal{E}$ /decoder $\mathcal{D}$), conditioned on two streams:

Geometric latent (concatenation). $v_{equi} = \mathcal{E}(V_{geo})$ is concatenated with the noised ground-truth latent $y_{equi,t}$ at each step — the 3D-aware geometric condition.

Image features (cross-attention). CLIP performs poorly on distorted ERP images, so the first frame I_0 is cropped into 8 perspective views (every 45° of yaw), each passed through a CLIP feature extractor $\mathcal{F}$; the resulting features are concatenated into c_{img} and injected via cross-attention.

2.5 Training Objective

Pantheon360 uses a standard diffusion denoising objective:

$$L = \mathbb{E}_{y_{equi}, v_{equi}, c_{img}, t, \epsilon} \left[\lambda(t) \|\epsilon - f_\theta(y_{equi,t}, t, v_{equi}, c_{img})\|_2^2 \right]$$

where $y_{equi} = \mathcal{E}(Y_{equi})$ is the ground-truth video latent, $y_{equi,t}$ its noised version at timestep t , $v_{equi} = \mathcal{E}(V_{geo})$ the geometric-scaffold latent, and c_{img} the semantic conditioning features. This jointly injects 3D geometric information and semantic context into the denoising process.

Implementation details. Both single-anchor and dual-anchor models are trained at 1024×512 resolution on 4 A100 GPUs for 5 days each. 3D reconstruction uses π^3 with a confidence threshold of 0.25 and sky masking.

2.6 Data Curation and Training-Pair Generation

Data source. The model is trained on **360-1M**, a large-scale collection of diverse, real-world 360° videos (not synthetic-only), after a filtering pipeline that removes mislabeled 180° videos, static posters, and low-motion clips.

On-the-fly annotation. Since 360-1M lacks ground-truth camera poses and 3D geometry, the authors auto-annotate training pairs (Y_{equi}, V_{geo}) on the fly: for each ground-truth clip Y_{GT} , they set $Y_{equi} = Y_{GT}$, then estimate 3D structure and camera trajectory C_{GT_poses} using **ViPE** (a video pose engine robust for 360° video), taking the resulting DROID-SLAM point cloud as the 3D Cache. High-quality, low-noise geometry is essential so the model learns to *trust* V_{geo} rather than learn to ignore it. Finally V_{geo} is rendered by setting $C_{target} = C_{GT_poses}$.

2.7 Dual-Anchor Latent Fusion for Interpolation

A dual-anchor variant is trained conditioned on both a start and an end frame, enabling interpolation between sparse observations (e.g., Google Street View panoramas). Because sparse-view point clouds can be geometrically inconsistent with the target end frame (causing jumps/discontinuities), the authors adopt the latent-fusion technique from *Time Reversal Fusion*, smoothly blending information from both anchor frames at the latent level. This is especially valuable for challenging real-world reconstruction scenarios such as Street View synthesis.

3 Experiments

Pantheon360 is evaluated across four regimes: single-view 360° generation, sparse-view 360° generation, two-view novel-view synthesis / interpolation, and comparison against 360° “world models.” Metrics: PSNR, SSIM, LPIPS, FVD (pixel-level quality) and MET3R (3D/multi-view geometric consistency); lower is better ($\downarrow$) except PSNR/SSIM ($\uparrow$).

3.1 Single 360° View-to-Video Generation

Setup. Baselines (ViewCrafter, TrajectoryCrafter, GEN3C), originally perspective-only, are adapted to 360° by cropping the rendered V_{geo} scaffold into 8 perspective views. Evaluated on **Web360** ($\sim 2,000$ in-the-wild outdoor clips; 100 sampled test sequences).

Table 1: Single 360° view-to-video generation on Web360.

Method	FVD $\downarrow$	SSIM $\uparrow$	PSNR $\uparrow$	LPIPS $\downarrow$	MET3R $\downarrow$
ViewCrafter	525.746	0.371	15.654	0.284	0.4914
TrajectoryCrafter	517.475	0.454	15.151	0.219	0.4578
GEN3C	380.080	0.583	20.730	0.145	0.3496
Pantheon360 (Ours)	356.151	0.746	22.838	0.065	0.2840

Pantheon360 outperforms all baselines on every metric, attributed to full panoramic FoV giving better cross-view consistency and a more complete scene understanding.

3.2 Sparse 360° Views-to-Video Generation

Setup. Multiple 360° keyframes at different timesteps; depth predicted per view via π^3 to build the 3D Cache. Evaluated on **Habitat** (34,000 synthetic indoor clips with non-looped, complex navigation polylines; 50 sampled sequences).

Table 2: Sparse 360° views-to-video generation on Habitat.

Method	FVD $\downarrow$	SSIM $\uparrow$	PSNR $\uparrow$	LPIPS $\downarrow$	MET3R $\downarrow$
ViewCrafter	778.207	0.193	11.833	0.398	0.5061
TrajectoryCrafter	690.322	0.216	12.223	0.461	0.6741
GEN3C	511.039	0.481	17.307	0.195	0.4522
Pantheon360 (Ours)	450.696	0.756	20.392	0.091	0.3026

Again best on all metrics, with a particularly large geometric-consistency gain (MET3R: 0.3026 vs. 0.4522 for GEN3C), confirming the diffusion model effectively follows the Cache’s geometric guidance.

3.3 Two-View 360° Novel View Synthesis

Only two 360° views are given (e.g., sparse Google Street View panoramas), and novel views are generated between them. Compared against **PanoSplatt3R**, Pantheon360 produces geometrically accurate interpolations with clean structure, while PanoSplatt3R shows visible artifacts and distortion.

3.4 Comparison with 360° World Models

Against **GenEx** (a high-level, action-controlled 360° world model) on Google Street View panoramas with simple forward motion, Pantheon360 maintains consistent quality and accurately follows the prescribed trajectory, while GenEx’s output quality degrades rapidly with accumulating geometric inconsistency.

3.5 Ablation Study: Dual-Anchor Latent Fusion

Four variants tested on 30 Google Street View scenes, measuring end-frame alignment (PSNR/SSIM/LPIPS), short-term warping error (STWE), and interpolation error (IE):

Table 3: Ablation on latent fusion for interpolation.

Method	STWE ↓	IE ↓	PSNR ↑	SSIM ↑	LPIPS ↓
Single	0.124	4.784	20.921	0.661	0.271
Single + Latent Fusion	0.420	12.083	28.006	0.817	–
Dual	–	–	27.86	–	–
Dual + Latent Fusion (Ours)	–	7.44	28.95	–	–

Note: The source table was truncated at the point of retrieval; the “Dual” and “Dual + Latent Fusion” rows are reconstructed from the surrounding text (which reports 27.86 PSNR for Dual and 28.95 PSNR / 7.44 IE for the full Dual + Latent Fusion method) rather than read directly from table cells, and some cells could not be confirmed.

Interpretation. The Single-frame model gives the best short-term temporal consistency but poor end-frame alignment (20.92 PSNR) since it has no signal about the target end state. Dual-anchor conditioning substantially improves convergence to the end frame (27.86 PSNR). The full method, **Dual + Latent Fusion**, achieves the best overall balance (28.95 PSNR, 7.44 IE), showing that latent fusion mitigates geometric inconsistencies from imperfect sparse-view point clouds while preserving smooth interpolation.

4 Applications

- **Video synthesis from sparse Street View data.** Pantheon360 generates consistent 360° videos from sparse Google Street View imagery, enabling smooth navigation across extended trajectories by chaining generated segments (the framework is the first video diffusion model supporting 360° interpolation for this purpose).
- **360° video stabilization.** Shaky 360° footage can be transformed into smooth, stabilized output by re-rendering along a smoothed trajectory through the 3D Cache.

5 Contributions (as stated by the authors)

1. Exact camera-trajectory control for *in-the-wild* 360° videos, overcoming prior methods restricted to simple action control or synthetic-only validation.
2. Pantheon360, a framework achieving this precise control via an explicit 3D Cache that enforces geometric consistency, freeing the diffusion model to focus solely on photorealistic texture refinement.
3. State-of-the-art 360° video synthesis performance across multiple tasks, with demonstrated utility in downstream applications (360° interpolation, video stabilization).

6 Discussion / Takeaways

The central architectural idea is a clean *separation of concerns*: an explicit, reconstruction-based 3D Cache handles geometric reasoning and camera-path fidelity, while a fine-tuned video diffusion model (SVD) is relieved of that burden and can specialize in photorealistic appearance synthesis. Combined with panoramic (360°) coverage rather than narrow perspective FoV, this substantially reduces the hallucination and cross-view drift problems that plague perspective-based camera-controlled video generation. The approach depends on the quality of the underlying 3D reconstruction (e.g., π^3 /VGGT at inference, ViPE/DROID-SLAM for training-pair generation); the dual-anchor latent-fusion mechanism is specifically introduced to compensate for cases where sparse-view reconstructions are imperfect.